Guards, Reputation and Prices: Testing Fixes for the Failures of Language-Model Agent Teams

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Abstract

Teams of language-model agents fail in predictable ways: they do not pool private information, they trust bad information, and, when trained with the wrong incentives, they stop cooperating. We test fixes for these failures in controlled, pre-registered experiments with Claude Haiku 5.5 teams, reporting each fix as the share of a gap closed against an explicit ceiling and pairing each mechanism with a prompt-only control. The strongest result is a defence against lying teammates that needs no ground truth: assigning a quarter of the honest agents as guards, who post, confirm and contest clues instead of solving, recovered 0.72 of the damage done by liars, against 0.34 for a track record computed from the true answers; their advantage held from 8 to 64 agents but shrank (from 0.26 to 0.17 in honest pooled share), because guards can only contest the clues they hold; splitting the team into squads of eight, each with its own guards, reversed that decline (the advantage grew from 0.26 at 16 agents to 0.34 at 64). The defence extended to mistaken and colluding teammates and to Claude agents prompted to deceive; but when the role was described and left unassigned, no agent took it, and training copies of a 3B model with team reward did not make it emerge. That track record itself, shown publicly, recovered a fifth to a third of liars' damage at every team size up to 32, while a private, costly check caught lies and recovered nothing; being public and being free each contribute about equally. Larger teams lose more to the same share of liars because agents build on a posted lie more often (from 0.79 of answers at 8 agents to 0.91 at 32), not because lies spread further. Mechanisms that add work to every agent's turn failed: a request board and an instruction to share first lowered output, and a board that displays how many agents agree added nothing over a one-sentence instruction. Training a 3B open model with reinforcement learning on the same task shows the incentive side: individual reward taught agents to guess instead of share, and pricing wrong answers suppressed guessing in every run, though its effect on sharing varied between training runs; training on teams of four and eight improved small teams but did not carry over to 32 agents. We release the environments, the frozen analysis plans and code that regenerates every number.

1 Introduction

Multi-agent deployments of language models are growing, from orchestrators with parallel workers [2] to large groups solving open-ended problems, and their failures are well documented. On hidden-profile tasks, where the right answer needs facts that only individual members hold, groups of agents reach about 30% accuracy where a single agent with all facts reaches 81% [12], and Anthropic's red team reports 17–36% for most models [3]. Agents conform [26], adopt a single false testimony readily [27], and become opportunistic under reinforcement learning in social dilemmas [9, 20]. The red team names the fixes as open problems: mechanism design, reputation, and environments that shape cooperation [3].

This paper tests fixes. For each failure we build a small, fully specified environment, freeze an analysis plan before any run, and add one mechanism at a time, reporting the *share of a gap closed* between the baseline team and an explicit ceiling. Each mechanism (an *affordance*: a new action, an observation line, a role) is paired with a prompt-only *instruction* control. Our contributions, strongest first:

1. **Verification as a role.** Guards, a quarter of the honest agents told to post, confirm and contest clues instead of solving, recover more of liars' damage than an oracle track record, with no ground truth; the advantage holds from 8 to 64 agents while shrinking, because guards hold only about two in five clues, and squads of eight with their own guards make it grow with team size instead; guards also help

against mistaken and colluding teammates and against Claude agents prompted to deceive; offered the role without an assignment, no agent took it, and team-reward RL did not make it emerge (§4.3).

2. **Public trust beats private verification, at every team size tested.** A harness-computed public track record recovers a fifth to a third of liars' damage up to 32 agents; private, costly checks catch lies but do not help; publicity and cost contribute about equally (§4.1–§4.2).

3. **Why larger teams are more gullible.** Exact loss decompositions of replayed runs show that agents in larger teams build on a posted lie more often; lies do not spread further (§4.2).

4. **Adding work to every turn fails.** When a message costs a turn, a request board and an instruction to share first lower output, and an agreement display adds nothing over an instruction (§5.1, §4.3).

5. **Incentives shape trained agents, less reliably than a few seeds suggest** (§6).

2 Related work

Information pooling in agent teams. The hidden-profile paradigm [23] has been ported to LLM groups [3, 12]. Diagnoses stress early convergence on shared evidence [12], averaging of expert and non-expert views [17] and loss of issue-critical facts in deliberation [24]; an LLM facilitator raised how much human groups shared without improving their decisions [1]. Adaptive communication control [4] and structured deliberation protocols [21] can beat unstructured discussion.

Communication cost. Pruning redundant messages and agents saves tokens at comparable accuracy [25, 29]; across 260 configurations, multi-agent architectures range from large gains to large losses against a single agent [11]. These measure cost in tokens; in our tasks a message costs task time.

Deception, trust and fault tolerance. Defences learn credibility scores from past contributions [6], rank neighbours [7] or rectify misleading content [15]. We compare a public, harness-computed signal with private verification at matched cost, and test oracle-free rules borrowed from social insects (quorum, stop signals, guards; 22). Honest-signalling theory predicts that verifiability matters [28]; our results suggest that who sees the verification, and whose job it is, matter too.

Training agent teams. Multi-agent RL for LLM collaboration includes MAPoRL [18], MAGRPO [14] and AT-GRPO [30]; counterfactual credit addresses free-riding [8, 13]. RL-trained LLM agents become opportunistic in social dilemmas [20] and most societies fail to sustain a commons [19]. Grading that never penalises wrong answers rewards guessing [10]; we observe the multi-agent analogue, a producer-scrounger outcome [5].

3 Setup

The clue board. N agents share a board of $2N + 2$ open subtasks, each worth 1, 2 or 3 points and open for 30 turns. A subtask hides an integer $0 \leq x < 30$ fixed by two clues "$x \bmod p = r$" dealt to different agents, so no agent can solve it alone. Each turn an agent opens a subtask, submits an answer (a wrong answer costs 0.5), shares a clue, or waits; shared clues go to a *blackboard* every agent reads. Welfare is points per agent-turn; the *pooled share* divides it by the welfare with every clue known to all, so 1 is the information ceiling. A scripted liar posts a false residue for a subtask it is not working on every other turn.

The hidden-profile vote. Four agents choose between two options; shared facts favour the wrong one and each agent holds two private facts favouring the right one, so only pooled facts reveal it. Agents post facts or opinions for three rounds, then vote; the ceiling is one agent given every fact.

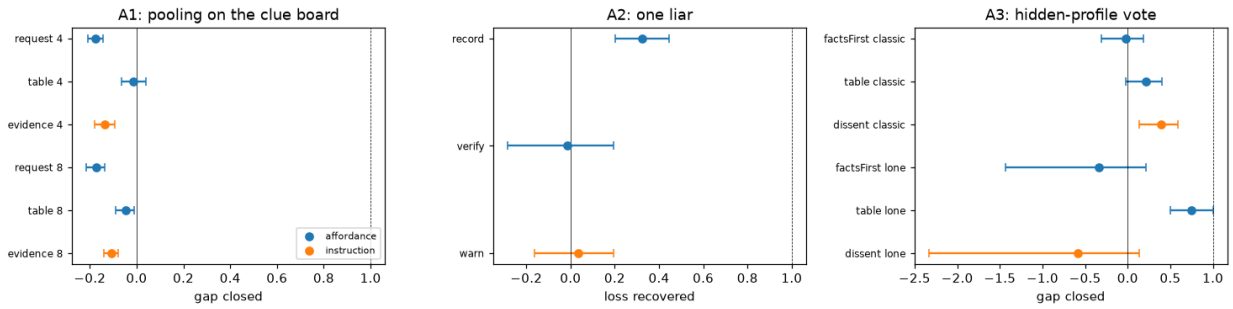


Figure 1: Share of the gap closed (pooling and vote) or of the liar's damage recovered (liar) per arm, four agents, with 95% intervals. Blue: affordances; orange: prompt-only instructions. 0 is the baseline, 1 the ceiling.

Agents and protocol. Claude Haiku 5.5 with low effort and structured JSON replies. Replies are cached, so every rerun is byte-identical. Each analysis plan was hashed and committed before its runs. Gap closed is the mean paired difference (arm minus baseline) over the ceiling minus the baseline; loss recovered is 1− (arm's mean honest loss / baseline's), where honest loss is the honest agents' welfare without liars minus with them. Intervals are 95% bootstrap over seeds or episodes; primary tests are Holm-corrected within each part [16]. All findings are exploratory.

4 Defending a team against liars

4.1 Public reputation, not private verification

One scripted liar among four cost the honest agents 0.35 welfare per agent-turn. Showing, next to each shared clue, how many of its sender's earlier shares were true (counted over subtasks already solved, so the record is never wrong) recovered 0.32 $[0.20, 0.44]$ of that loss (Holm $p = 0.008$), with no cost when no liar was present (0.006 $[-0.012, 0.026]$). A *check* action that let an agent spend a turn to learn whether a posted clue was true was used 14 times per board and caught 13 false clues, yet recovered nothing (-0.01 $[-0.28, 0.19]$); a warning instruction did nothing (0.03 $[-0.16, 0.20]$) (Figure 1).

To separate the two ways the record and the check differ, a private record and a public check completed a 2×2 of public or private and free or costly (10 seeds). Pooled over the design, publicity added 0.18 $[0.04, 0.36]$ to the loss recovered and being free 0.16 $[0.02, 0.30]$ (difference 0.02 $[-0.16, 0.20]$): both matter, about equally. (For the record pair, privacy and the amount of information are confounded, as each agent sees only its own solved subtasks; the check pair is not.)

4.2 Does it scale, and why do larger teams lose more?

With 8, 16 and 32 agents and a quarter of them liars (5 seeds; plan frozen first, the 32-agent cost checked in a pilot), the track record recovered 0.30, 0.22 and 0.31 of the liars' damage; every bootstrap interval excludes zero, though with five seeds exact sign-flip tests cannot fall below $p = 0.0625$. Larger teams lost more to the same share of liars (honest loss 0.47 at 8 agents, 0.54 at 32).

We replayed these runs from the response cache with a turn-by-turn tracer (a further frozen plan; no API calls) and split the honest loss into additive terms that sum to it exactly. Lies did not become more visible in larger teams, and agents almost never passed a liar's clue on (at most 0.25% of liar posts). What grew was how often agents built on a lie: the share of answers resting on a false clue rose from 0.79 at 8 agents to 0.91 at 32, such answers were wrong 95–97% of the time, and the loss on subtasks a lie touched carried 0.72 of the total rise (interval $[0.16, 3.2]$). Once a liar had posted a clue slot, the true clue was never posted afterwards in 89–94% of cases: agents treat a posted slot as settled, whoever posted it.

4.3 Guards: verification as a role

The track record needs the harness to know the truth. Taking the settled-slot failure and social insects as the cue (colonies use redundancy, quorum, stop signals and guard castes), we tested oracle-free mechanisms with each clue held by two agents and a quarter of the agents liars, at 8 and 16 agents (8 seeds; plan frozen first, `58e842277a3e`; $24.89): a board that counts how many agents posted each value and marks values confirmed (two senders) or contested; the same advice as a one-sentence instruction, without the display; and the board with a quarter of the honest agents assigned as *guards*, told to post, confirm and contest clues and never to solve.

Guards worked. They beat the counting board by 0.155 [0.135, 0.183] in honest pooled share with liars (Holm p <0.001) and the unprotected board by 0.26 at 8 agents and 0.20 at 16 (exact $p = 0.004$ at each size), recovering 0.72 of the liars' damage, against 0.34 for the oracle track record. They did so by concentrating checking: guards contested 0.31 false values per lie (counting board alone 0.21, plain board 0.06), and the share of honest answers built on a lie fell to 0.31 (counting board 0.60, plain board 0.73). **The display did not.** The counting board helped (0.075 [0.048, 0.102], Holm p 0.013), but no more than the one-sentence instruction (−0.001), and both faded from 0.12 at 8 agents to 0.03 at 16. Every arm cost about 0.07 pooled share on clean boards, and some honest agents reposted a lie to "confirm" it, a known weakness of counting agreement among agents who relay. The guards' advantage fell from 8 to 16 agents by −0.06 per seed (two-sided exact $p = 0.17$, not significant), which under the frozen rule stopped the planned test at 32 and 64 agents.

Guards at 32 agents. Because that decline was not significant, we ran one more size as a declared deviation, with its own plan frozen first (`f51f0298d34a`; $32.73): the plain board and guards at 32 agents, clean and with each of three kinds of faulty member. With independent liars, guards beat the plain board by +0.229 [0.182, 0.274] (exact $p = 0.004$, all eight seeds); across 8, 16 and 32 agents the advantage showed no trend (−0.016 per doubling, two-sided $p = 0.26$). At this size liars wiped out the unprotected team (honest pooled share −0.08 against 0.36 clean; 0.85 of honest answers built on a lie), and guards closed 0.52 of that gap (contamination 0.45). Guards also helped against *mistaken* teammates, who misread 30% of the clues they hold without intent (+0.058 [0.024, 0.094], $p = 0.023$), and against *colluding* liars, who back each other's false values and so defeat a two-sender quorum (+0.090 [0.054, 0.118], $p = 0.008$): a guard contests a wrong value of a clue it holds however many agents posted it. The clean-board cost at 32 agents was 0.043.

Guards at 64 agents, and why their advantage shrinks. A second declared deviation (plan frozen first, `9764cd43e34c`; $27.04) ran the plain board and guards with independent liars at 64 agents. Guards still won by +0.165 [0.142, 0.185] (exact $p = 0.004$, all eight seeds): liars pushed the unprotected team below zero (−0.09; 0.85 of honest answers built on a lie) while guards kept 0.08 (contamination 0.55). Over 8 to 64 agents, however, the advantage fell by −0.026 per doubling (two-sided $p = 0.016$; descriptive). Replaying the 64-agent boards from the response cache at no cost showed why. Guards did not weaken: on clues they held they contested 0.75 of lies, usually within a round, and each such lie led to 0.24 wrong answers. But with two holders per clue and a quarter of the team guarding, only 0.37 of lies landed on a clue a guard held; the rest led to 1.49 wrong answers each, more in larger teams, and caused 0.91 of all answers built on a lie. The limit is coverage, not the guards.

Guards that scale. We then tested three design changes against these guards at 16 and 64 agents (plan and numpy forecast frozen first, `a441381ce078`; $43.80). All three beat them at 64 agents (Holm $p = 0.012$ each, all eight seeds, and on raw welfare too). *Squads*, eight agents per board with two guards each and their own subtasks, won by +0.171 [0.156, 0.186], and their advantage over the unprotected team *grew* with size, from 0.26 at 16 agents to 0.34 at 64 (difference-in-differences against guards +0.111 [0.069, 0.152]); a 64-agent team of squads kept 0.25 honest pooled share, near the 8-agent guard board's 0.28, with prompts of 780 tokens instead of 6,463. *Failure flags*, which mark on the board every posted value that agreed with an answer that failed (a signal the environment already gives the submitter), added +0.062 [0.042, 0.082] at 64 agents and +0.054 at 16, at no clean cost (+0.000), though 0.27 of flags fell on true values. *Guard copies*, one copy of every clue dealt to a guard, won only at 64 agents (+0.043 [0.025, 0.063]): they cost −0.129 on clean boards at 16, and the guards, now responsible for every clue, contested 0.61 of lies against 0.75. Verification has a capacity limit, and small groups with their own checkers scale where one global board does not.

Claude liars and unassigned roles. Two further tests at 16 agents (plan frozen first, `0c59fa10f2b0`;

$8.41) replaced the scripted liars with Claude agents told, in secret, to make teammates submit wrong answers, and offered the guard role to every honest agent without assigning it. Guards beat the plain board against the Claude liars too (+0.063 [0.016, 0.104], Holm $p = 0.039$), recovering 0.70 of their damage (0.63 for scripted liars) and cutting answers built on a lie from 0.39 to 0.18. The Claude liars lied less often than scripted ones (an accepted false clue in 0.18 of their turns, against 0.5), each lie did the same damage (0.006 against 0.006 per lie), they did not avoid clues a guard held (0.38 of their lies), and, if anything, lied more often when guards were present (0.25 of turns). Offered the role with its rationale but no assignment, no honest agent acted as a guard on any board: the team did no better than the counting board alone (+0.009 [−0.023, 0.039], Holm $p = 0.293$), −0.16 against assigned guards, and the added instruction cost 0.10 on clean boards. The division of labour paid only when it was assigned.

Can training make the roles emerge? We trained copies of one 3B model (one shared adapter) with team reward on the counting board, with no role instruction: five runs of 30 steps (plan frozen first, `c103584f5aac`; Phase 17's recipe with the per-token KL capped, after four of its halts proved to be outliers in the logged KL). Training was stable, but no division of labour emerged. Specialization above a shuffled-slot null (+1.32, exact $p = 0.0625$) was not significant, and the untrained warm start was already about as specialized (+3.44) as the most specialized trained run (+3.57): training kept, weakened or erased the structure it started with, never added to it. 2 of the five runs collapsed to every agent posting and none solving, because under team reward a post is never penalized and an answer can be. Neither prompting nor this reward produced the roles that assignment provided.

5 Pooling and aggregation

5.1 Communication has a price

The baseline team earned a pooled share of 0.45 at four agents. A *request board*, on which an agent posts that it needs clues for its open subtask, was used in about a quarter of turns (42 asks per board), sharing did not rise, and the gap widened: −0.18 [−0.21, −0.14] at four agents, −0.17 [−0.21, −0.14] at eight (Holm $p = 0.008$), and about the same at 16 and 32 (−0.17, −0.16). An instruction to post every held clue first nearly doubled sharing (0.40 to 0.73 shares per held clue) and also lowered output (−0.14 [−0.18, −0.09]). Sharing costs a turn here, and the extra communication spent turns that solving needed.

5.2 A vote fix that did not replicate

In our hidden-profile vote, members posted about half of their private facts (0.50) in every arm, yet a table of the facts posted so far closed 0.75 [0.50, 1.00] of the gap when one member held a decisive fact (win rate 0.88 to 0.97; Holm $p = 0.012$), mainly by changing votes. Rerun unchanged on HiddenBench's 65 tasks [12] (195 episodes per arm, plan frozen after a pilot), the table did not help (−0.07 [−0.69, 0.30] of the gap); the one arm that moved was an instruction to speak up, which raised disclosure of hidden facts from 0.74 to 0.88 and the win rate from 0.77 to 0.83. Our protocol left little headroom (agent accuracy 0.79 against the benchmark's reported 30%; solo 0.91), and we no longer claim the aggregation effect in general.

6 Incentives when training a team

We trained Qwen2.5-3B-Instruct to play the clue board in teams of four copies sharing one LoRA adapter: a warm start imitating a rule agent, then 60 steps of group-relative policy gradient with a KL penalty to the warm start and a learning rate steered to a target per-step policy change (Figure 2). Imitation did the teaching: welfare rose from −0.001 to 0.27 [0.22, 0.32]. The reward decided what RL taught on top. Under *individual* reward (an agent's own solves) the model stopped sharing (0.92 to 0.15 shares per held clue) and solved by guessing (74% of solves): with a wrong answer costing 0.5, guessing among three candidates for a 3-point subtask earns about 1.25 per turn, more than sharing pays the sharer. Copies with identical weights became selfish because the reward, not the weights, defines their interests. *Team* reward (every solve shared) was unstable; *credit* reward (the solver keeps half, the rest goes to the agents whose clues it used) kept sharing and beat individual by 0.19 [0.09, 0.27].

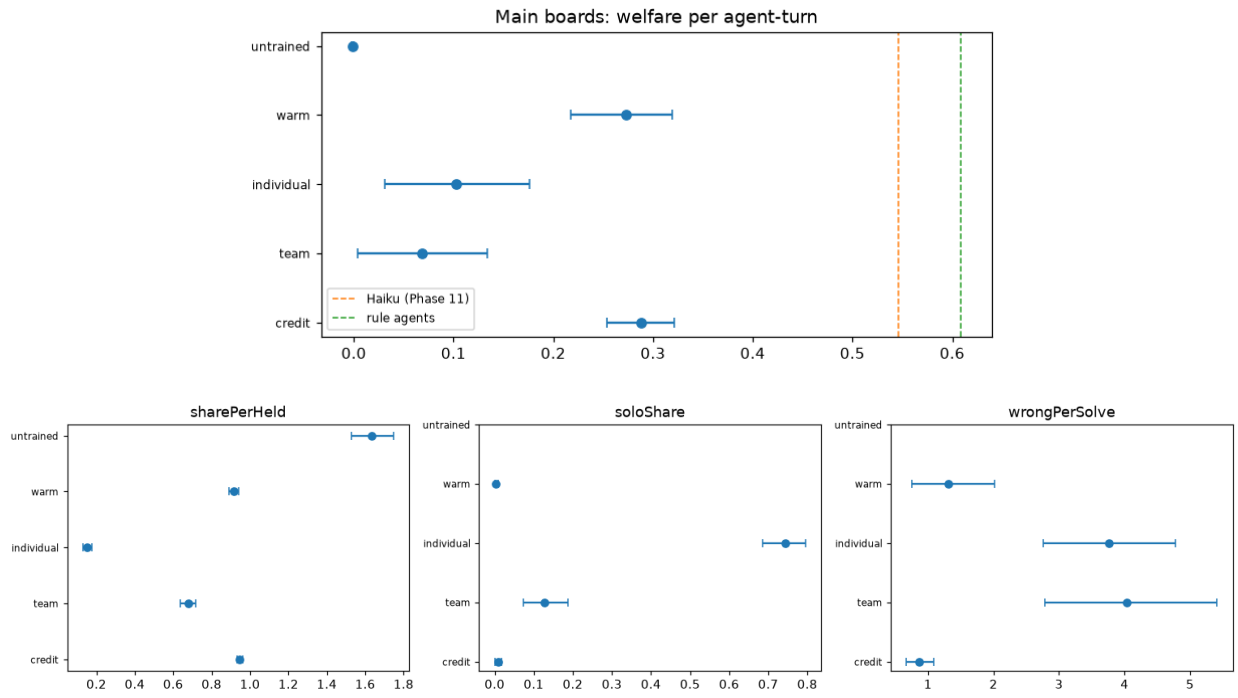


Figure 2: Top: welfare per agent-turn on held-out boards (Haiku and rule agents for reference). Bottom: shares per held clue, share of solves by guessing, wrong answers per solve.

Pricing the loophole, and the training run as the unit. A training-only cost of 3 per wrong answer under individual reward (`indpen`) suppressed guessing in every training run (0.1–15 guesses per board against 49–90 without it). With two seeds it also looked as if it kept agents sharing; a pre-registered replication with four unpriced and five priced runs, tested with exact permutation tests over runs, did not confirm that: priced runs shared more on average (+0.37 clues per held clue $[0.04, 0.67]$, exact $p = 0.056$, Holm 0.11), but individual runs ranged from 0.10 to 1.00, and the selfish drift appeared in only 2 of the four unpriced runs. Gains over imitation were seed-dependent too: one credit configuration gained 0.02 with one seed and 0.12 with another. An environment fix reliably changed the behaviour it targeted and changed the team outcome less reliably, and one or two training runs per design are not enough to rank reward designs.

Training for team size. Trained 3B teams lose most of their pooling as teams grow, as Claude teams do far less. Eight runs of 60 steps on a mix of four- and eight-agent teams with liars (plan frozen first, `8e28b5fdfb1f`; the training run as the unit) raised welfare at four agents from 0.26 to 0.34 (+0.078 $[0.008, 0.136]$, exact $p = 0.035$, 7 of 8 runs positive, Holm 0.105 over three primaries), but the gain shrank with size (-0.072 $[-0.140, -0.012]$ from 4 to 32 agents), and at 32 agents welfare stayed at 0.09 against 0.08 for the warm start and 0.44 for Claude Haiku; pooling did not rise (+0.006). Trained teams did lose less to liars at 32 agents (secondary, +0.061 $[0.012, 0.108]$, $p = 0.023$). Five of the eight runs halted early, four of them on one-step outliers of the logged KL to the warm start (the per-token estimate is unbounded when the trainer scores a sampled token far below the reference) rather than a change in behaviour. Training at small sizes did not teach what large teams need; in our runs, the structure of the team did.

7 Discussion

Where to put the work. The mechanisms that helped made information easier to trust at no cost in turns (a public track record) or gave checking to agents whose job it was (guards); the ones that failed added work to every agent's turn (asking, checking, sharing first) or only displayed information agents could

already infer (agreement counts). This suggests a design principle for agent harnesses: put verification in the environment or in a role, not in every agent's turn. Guards are the oracle-free version of the principle, a division of labour among copies of one model. Prompted agents did not adopt it on their own even when told its value, so the natural next question is whether a team trained with a shared reward discovers it without being told.

Trust as a public good. A private check produced information worth exactly one agent's next action; a public record produced it for everyone. In our earlier numpy simulations of a liar in a foraging world, private trust weighting was the defence that paid, so the public/private contrast for LLM agents is not a foregone conclusion.

Limitations. The tasks are small and synthetic; the Claude teams use one model (Haiku 5.5) and scripted liars, not adaptive ones; board arms have 5–10 seeds and the RL arms one to eight training runs, which our own results show is too few to rank similar designs; RL used a 3B model trained at four and eight agents only. Guards were tested at 8 to 64 agents, against scripted liars and mistaken teammates; squads had no subtasks spanning squads, so their gain is an upper bound for tasks that split cleanly, and the design changes were tested against scripted liars only; Claude liars could act only through clue posts (no free-text persuasion), and correlated errors shared by every copy of a model and compromised agents remain untested; the 32- and 64-agent tests were declared deviations from the frozen plan; Stage 4 ran in two seed batches, analysed only as a whole. The trust fixes have not been tested on an established misinformation benchmark [27].

Broader impact. The fixes studied are defensive (against misinformation) or improve cooperation; the RL result shows how easily a plausible incentive teaches agents to abandon cooperation, which matters for anyone training groups of models.

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A Reproducibility

Code, environments and notebooks: https://github.com/Positron00/explRL. Analysis plans were frozen and committed before their runs (fixes: `FIX_PLAN`, hash `79abfb531cdd`; RL: `RL_PLAN` v2, hash `4d1ea0078a6e`; second RL round: `RL13_PLAN`, hash `3e50a4002c7b`; HiddenBench: `HB_PLAN`, hash `791a2df079a8`, frozen after a pilot, on the benchmark's 65 tasks vendored unchanged under their MIT licence; trust information: `TRUST_PLAN`, hash `6d2818ae84c2`; team size: `SCALE_PLAN`, hash `b886baefb2ef`, frozen after a pilot; traces: `TRACE_PLAN`, hash `f5c25637acd0`; guards: `COLONY_PLAN`, hash `58e842277a3e`; guards at 32 agents: `COLONY32_PLAN`, hash `f51f0298d34a`; guards at 64 agents: `COLONY64_PLAN`, hash `9764cd43e34c`; guards that scale: `STAGE4_PLAN`, hash `a441381ce078`; trained roles: `RL18_PLAN`, hash `c103584f5aac`; team-size RL: `RL17_PLAN`, hash `8e28b5fdfb1f`; Claude liars and unassigned roles: `COLONY3_PLAN`, hash `0c59fa10f2b0`). All API replies are cached locally (available on request), and cached reruns reproduce every row and turn log byte for byte at no cost. Every number in this paper is generated from the saved results by `paper/make_assets.py`.

B A cheap simulation as a screen

Rule-agent simulations in numpy ranked harness designs the way Haiku teams did on a task board (Spearman $\rho = 0.75$ $[0.55, 0.93]$) and the clue board ($\rho = 0.79$ $[0.74, 0.81]$), but not on text foraging ($\rho = 0.02$ $[-0.47, 0.45]$). Forecasts frozen before the expensive runs were right more often than not but missed the most interesting effects (pricing the loophole; guards); we study them as a design-screening method in a companion paper.